Aryan Naik

aryannaik1205@gmail.com | github.com/RobotsAndAryan

EDUCATION

MEng Robotics and Artificial Intelligence | University College London (UCL) | 2025 - 2029

TECHNICAL SKILLS

Robotics and Automation: Arduino microcontrollers, LiDAR spatial mapping, PWM actuator control, kinematic control loops

Programming: C++, embedded systems logic

CAD: Autodesk Fusion (component geometry, cell layout)

Electronics: Breadboard circuit design, hardware integration, functional circuit testing

PROJECTS

Autonomous Terraforming Robot

- Delivered reliable obstacle avoidance across a mapped environment, demonstrated by successful LiDAR-based spatial mapping and zero collision incidents during testing, by implementing C++ kinematic control loops on an Arduino GIGA R1 dual-core microcontroller
- Achieved precise actuator response across all motion axes, confirmed by consistent PWM signal output under load, by designing and integrating hardware PWM actuator control into the robot drive system

Robotics Challenge

- Produced a fully integrated electronics hardware assembly, confirmed by all functional circuit tests passing to specification across every tested component, by designing and wiring breadboard circuits with microcontroller logic for a multi-component robotic system
- Delivered a manufacturable component design meeting all dimensional requirements, verified by design review sign-off against project tolerances, by applying CAD modelling in Autodesk Fusion to develop physical robot geometry and cell layout

KEY SKILLS

- Analytical and fault-finding problem solving applied across hardware and software integration
- Practical, hands-on approach to engineering challenges demonstrated through project build and test cycles
- Strong organisational ability managing concurrent project deliverables across design, build and verification phases
- Effective communication and stakeholder engagement developed through academic project presentations and team reviews
- Genuine interest in robotics, automation and new technology in practical engineering environments

ADDITIONAL INFORMATION

- Actively developing knowledge of PLC systems, Lean Manufacturing methodologies, Six Sigma and Statistical Process Control through self-directed study aligned to industrial automation practice
- Familiar with APQP and PFMEA frameworks as part of MEng coursework in manufacturing systems and quality engineering
- Available for internship from Summer 2026

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Hiring Manager Stirling Warrington

Dear Hiring Manager,

I am writing to apply for the Graduate Automation Engineer role at the well-established UK manufacturing business you represent. The emphasis your client places on investing in robotics, PLC systems and capital equipment to modernise a complex production environment is precisely the kind of work I am eager to contribute to. The opportunity to deliver measurable improvements in efficiency and quality, rather than simply observe from the sidelines, makes this role stand out as an exceptional early-career placement.

My most directly relevant project involved designing and building an autonomous terraforming robot from the ground up. I wrote C++ kinematic control loops running on an Arduino GIGA R1 dual-core microcontroller, integrated LiDAR for real-time obstacle avoidance and spatial mapping, and implemented hardware PWM actuator control to achieve precise, reliable movement. The result was a fully functional autonomous system capable of navigating and operating in unstructured environments, demonstrating my ability to take a robotics project from concept through to working hardware. Alongside this, a separate electronics challenge gave me hands-on experience with breadboard circuit wiring, microcontroller logic and CAD modelling in Autodesk Fusion, reinforcing my practical approach to engineering problems.

In this role I would contribute a strong foundation in embedded systems, autonomous control and hardware integration, and I am confident I can apply these skills to support the commissioning and optimisation of robotic and automated systems on the production floor. I am keen to develop my understanding of PLC-controlled processes, Lean manufacturing methodologies and industrial automation within a real manufacturing context, and I believe this placement would provide an outstanding environment in which to grow as an engineer.

Yours sincerely, Aryan Naik